VRUSHALI RAJESH SARWAN

Pune, MH, India | +91 9156802406 | vrushalisarwan30@gmail.com | LinkedIn

PRODUCT OWNER & HEAD OF CODING / PROGRAM LEADER — K-12 STEM & CODING EDUCATION

End-to-End Product Ownership & K-12 Coding Platform Roadmap | AI-Native Learning Layers (AI Tutors, Automated Code Review & Grading) | Agile Backlog Prioritization & Sprint Governance | K-12 Project-Based Learning (PBL) & Scratch/Python Pedagogies | Youth Hackathons & STEM Showcase Operations | Student Outcome Measurement & Metric Tracking

EXECUTIVE SUMMARY

Visionary Product Owner and Program Leader for K-12 STEM & Coding Education with 8+ years of experience leading end-to-end product lifecycles, technical curriculum architecture, and multi-cohort learning delivery across independent and institutional platforms. Combines an IISER Pune analytical background (BS-MS Dual Degree) with expertise in defining feature roadmaps, prioritizing engineering backlogs, and integrating AI-native learning layers—including interactive AI tutors, automated code feedback engines, and real-time evaluation rubrics. Project Lead for the prestigious Atal New India Challenge (AIM, NITI Aayog), orchestrating nationwide youth hackathons, STEM showcases, and project-based learning competitions. Proven track record in designing scalable Scratch, block-based, and Python learning paths, mentoring instructor networks, and managing student outcome metrics to maximize engagement, retention, and competitive performance.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Product Ownership & K-12 Platform Roadmap:** End-to-End Product Lifecycle Management, Feature Roadmap Definition, User Story Writing, Agile/Scrum Backlog Prioritization, Feature Flagging, Product Requirements Documents (PRDs), Wireframing & UX Benchmarking.
- **AI-Native Learning & Automated Evaluation Tools:** AI-Native Learning Layers, Conversational AI Tutors, Automated Code Review & Linting Engines, Generative AI Prompt Engineering, Real-Time Assessment Rubrics, Automated Grading & Feedback Pipelines.
- **K-12 Coding Pedagogies & Curriculum Design:** K-12 Project-Based Learning (PBL), Block-Based Coding (Scratch/Blockly), Python & Computational Thinking Frameworks, Modular Curriculum Architecture, Instructor Onboarding & Mentorship.
- **Hackathons, Competitions & Student Showcase Operations:** Youth Hackathon Orchestration, STEM Science Exhibitions, Live Technical Demonstrations, Judging Criteria & Scoring Rubrics, Student Pitch Coaching, Event Logistics Management.
- **Outcome Measurement & Platform Analytics:** Learner Skill Progression Metrics, Cohort Retention Analytics, Power BI & Tableau Dashboards, Python Data Analytics, SQL, Automated Certification engines, SLA & ROI Tracking.

CHRONOLOGICAL PROFESSIONAL EXPERIENCE

1. Senior Facilitator (Business Analytics Track)

Regenesys Business School | January 2025 – Present

- **Product Ownership & Learning Platform Roadmaps:** Spearhead platform feature roadmap planning and agile user story creation in close collaboration with engineering teams. Manage sprint backlogs to deploy automated feedback tools and platform enhancements, raising feature adoption by 35% and delivery SLA compliance to 98%.
- **AI Tutor Integration & Quality Evaluation:** Architect AI-driven learning tools and interactive evaluation rubrics. Standardize facilitator delivery, conduct systematic audits, and implement automated performance feedback for 500+ active learners across parallel tracks.
- **Product Backlog Prioritization & Outcome Metrics:** Define product metrics for tracking skill progression, ramp intervals, and program retention. Prioritize bug fixes and user feedback tickets, cutting administrative workflow overhead by 40% while maintaining 99.8% system stability.
- **Stakeholder Alignment (CSM & IBO Networks):** Partner closely with Customer Success Managers (CSMs), enterprise clients, and Independent Business Owners (IBOs) to harmonize delivery targets with strategic business outcomes. Synthesize complex learner trends into executive Power BI dashboards to validate learning ROI.

2. Senior Data Science Faculty

Fusion Software Institute, Pune | July 2024 – January 2025

- **Learning Delivery Oversight & Vendor Management:** Orchestrated multi-week corporate bootcamps and tech training programs, supervising third-party instructor sourcing, facilitator onboarding, and cohort scheduling for 300+ working professionals.
- **Quality Audits & SLA Performance Tracking:** Instituted regular virtual classroom audits and instructor scoring rubrics based on Phillips evaluation frameworks. Monitored program SLA metrics, ramp interval milestones, and customer success targets, boosting course completion rates by 22%.
- **Cross-Functional Client Success Alignment:** Collaborated with account teams, CSMs, and business leads to analyze learner engagement trends, optimize delivery workflows, and deliver structured status reports on key performance indicators.

3. Data Science Faculty & Associate Trainer

ITVedant, Pune | October 2022 – June 2024

- **Virtual Learning Environment & Platform Operations:** Governed LMS operations, virtual lab infrastructure, and digital learning platforms for 250+ active learners across parallel tracks, ensuring zero platform downtime and strict SLA adherence.

- **Learner Analytics & Ramp Interval Monitoring:** Tracked participant progression, assignment completion rates, and ramp intervals using Power BI and automated tracking sheets. Spearheaded proactive intervention programs for struggling cohorts, raising overall certification rates to over 95%.
- **Instructor Evaluation & Standards Alignment:** Executed routine instructional quality reviews, standardized grading rubrics, and facilitated continuous feedback sessions with internal trainers to enforce delivery excellence.

4. Data Science Faculty (Academics & Operations)

Ajeenkya DY Patil University, Pune | August 2022 – October 2022

- **Delivery Standards & Quality Engineering:** Directed course delivery for undergraduate and MBA cohorts. Standardized lesson delivery blueprints, learning objectives, and Phillips-inspired evaluation criteria, resulting in a 20% increase in pass rates.
- **Multi-Cohort Operations & Resource Planning:** Managed schedule alignment, faculty resource allocation, and LMS content deployment, maintaining operational compliance with institutional SLAs.

5. Data Science Faculty & Postgraduate Mentor

MKSSS'S AIT, Center for Data Science, Machine Learning and AI, Pune | April 2021 – August 2022

- **L&D Infrastructure Governance & SLA Tracking:** Managed operational setup and technical governance for BI computing platforms (Power BI, Tableau, RStudio). Reduced setup turnaround times by 50% and instituted standard operational maintenance SLAs.
- **Mentorship, Quality Audits & Certification Metrics:** Guided 200+ scholars in data analytics projects, conducted quality evaluations on research outputs, and managed certification verification processes to achieve a 98% program completion rate.

6. Head of Coding & STEM Program Leader (Independent Practice)

Independent Educational Coaching Practice | July 2018 – March 2021

- **K-12 STEM & Coding Product Ownership & PBL Curriculum Architecture:** Founded and directed an independent STEM & coding practice, managing end-to-end learning product delivery for 150+ K-12 students. Designed progressive project-based learning (PBL) pathways spanning Scratch, block coding, and foundational Python, incorporating automated code review workflows and real-time skill progression tracking.
- **Youth Hackathons & Competition Leadership (Atal New India Challenge):** Served as Project Lead for the Atal New India Challenge (Atal Innovation Mission, NITI Aayog, Govt of India, 2018). Orchestrated nationwide student hackathons, science exhibitions, and tech showcase events, managing judging rubrics, project evaluation workflows, and cross-functional stakeholder logistics for youth innovators.

KEY TECHNICAL & APPLIED PROJECTS

- **AI-Native Automated Code Review & Grading Engine:** Designed and deployed an automated code submission, linting, and grading module integrated into student learning platforms, providing instant syntax feedback and error explanations for Python and block coding learners.
- **K-12 Hackathon & Project Showcase Platform Engine:** Built a digital submission and evaluation portal for youth STEM competitions and hackathons, implementing standardized scoring rubrics, peer review systems, and automated certificate generation.
- **Conversational AI Systems & Executive NLP Tools:** Engineered Rasa Chatbot architectures integrated with Flask RESTful APIs. Built automated SQL query engines and Excel Intelligence Chatbots, enabling stakeholders to retrieve analytical insights via natural language with 92% accuracy.
- **Earthquake Magnitude Statistical Thesis (IISER Pune):** Formulated statistical methods to estimate magnitude completeness (Mc) and spatial uncertainties using synthetic data platforms, stochastic simulation algorithms, and custom MATLAB scripts for seismic hazard assessment frameworks.

CERTIFICATIONS & TECHNICAL CREDENTIALS

- **Honors & Credentials:** Phillips ROI Methodology Practitioner | Certified in AI Delivery & Deployment (ITVedant) | 'Employee of the Month' Enterprise Award (ITVedant) | IBM Professional Certifications in Machine Learning & Data Analysis | ML Engineer - AI Analyst Certification (Symbiosis Skills) | Advanced Generative AI & Prompt Engineering Specialist.
- **Publications & Research:** Co-author, "Machine Learning in Supply Chain and Analytics" (Phygital) | BS-MS Master's Thesis on Earthquake Magnitude Completeness (IISER Pune) | Author, IEEE Research Paper "HyPE: Positional Encodings for Length Generalization in Transformers".

EDUCATION & ACADEMIC PEDIGREE

- **Degree & Institution:** Integrated Masters (BSMS Dual Degree) in Mathematics and Earth & Climate Science | First Class | Indian Institute of Science Education and Research (IISER), Pune.
- **Fellowships & Awards:** National INSPIRE Fellowship Recipient (DST, Govt of India, 2019) awarded to top 1% meritorious students nationwide | Pune Municipal Corporation Academic Scholar (2012, 2014) in recognition of scholastic achievement.
